

Data Safety and Recovery

(45 theory lessons, 30 practice lessons)

CHAPTER 5: DATA PROTECTION MECHANISMS & SPECIALIZED FILE SYSTEM



Contents

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- Establish specialized file system
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Build data protection mechanisms

Using available tools

- System login tools (on OS/ firmware, computer, hard disk drive, etc.)
- Data encryption tools (transfer data to encrypted files and check on passwords).
- Hide data inside files
- Firewalls and anti-virus software.
- Update system

Authentication with a password and dynamic key

- To avoid the passwords being guessed, strong and unique passwords should be used.
- Password manager tools should be considered to use. They can create and store the safe passwords.

Authentication with a password and dynamic key

- Besides, there is also an option to design a system with constantly changing passwords. The password authentication algorithm can be based on the system clock or specific contexts.
- Stock trading, e-commerce, money transfer, etc.. a similar 2-factor authentication approach is used: in addition to the common password, users are also asked to enter an OTP code sent via SMS to their registered phone number or taken from Token card devices.

Authentication with a password and dynamic key

- The soft key is a method that helps to protect data using software tools (it's also used in intellectual property rights protection including copyrighted software). Conversely, the hard key method (secure dongle, hardware key, hardlock, USB dongle, etc.) helps to protect data using hardware devices.

Authentication with a password and dynamic key

- Each method has its own pros&cons, and they work as a complement to each other. Although the concept of the soft key was brought up by developers the earliest, the hard key method was the first one to be actually invented and used. The hard keys are fairly easy to manufacture and use. At a simple level, developers can even utilize USB flash as a hard key by designing a range of specific programs/ structures for it.

Encryption with symmetric/asymmetric key algorithms

- Symmetric encryption (non-public key):
 - Encryption is considered as a symmetric algorithm when encrypted keys and decrypted keys expose obvious relations with each other (a.k.a easy to figure out a key when knowing the other). Encrypted keys and decrypted keys can be either totally similar to each other, or vary according to variations between them.
 - This algorithm is suitable for individual data self-encryption without the need of sharing keys, or with whom the decrypted keys can be passed over a secure channel. In case of not having an assurance for keys exchange, asymmetric encryption (public key) would be a good choice.

Encryption with symmetric/asymmetric key algorithms

- Asymmetric encryption (public key):
 - This algorithm uses a pair of keys (k_1 & k_2). k_1 is used for encryption and k_2 is used for decryption, and vice versa. Either of them can be private (a.k.a Private key) and the other will be public (a.k.a Public key). Although the complication and data processing volume of asymmetric encryption is far bigger than the symmetric, this algorithm works effectively in situations of confidential data exchange.

Encryption with symmetric/asymmetric key algorithms

- Hard/SSD drive encryption:
 - Windows: BitLocker.
 - macOS: FileVault, Disk Utility
- Important notices:
 - Remember the encryption passwords,
 - Store files of recovery keys carefully (BitLocker)
 - Data is completely lost if both of them are lost
- Encrypt files, folders: Winzip, Winrar, 7-zip, ...

Steganography

- The art and science of concealing the existence of a message or data within another carrier object, typically a digital file, thus making the presence of the message undetectable.
- The goal of Steganography is to conceal the existence of the data, as opposed to Cryptography, which aims to conceal the content of the data.

Steganography

- Technique:
 - Modifying the redundant/least significant bits (LSB) of the carrier file without making a substantial or perceptible difference to the naked eye or ear. Examples include images and audio
 - Utilizing the redundant data areas of compressed files (such as JPEG, ZIP) or regions reserved for headers and comment fields to conceal the message.

Establish specialized systems

- If there is a need to store a lot of confidential data, a specialized file system can be counted. This system has a different structure compared to the current ones and does not publish the retrieval tools. Hence, the files can only be retrieved while in the system. Even if the whole volume and storage devices are stolen and taken away, the stealer will not be able to decrypt the files. The list of actual files and their equivalent contents are undefinable.

Establish a specialized file system

Introduction

When a specialized file system with a self-proposed structure is established, it means that the existing retrieval tools can not be applied and that no one can retrieve the data stored in the system (supposing encryption has been done and the design is not too simple/ too similar to an existing design).

You don't have to design a real file system like OS which requires support for direct access on the storage devices (mostly sector access), but you can build up a file system inside a particular file, such as .ZIP,.RAR,.ISO, etc. In a more complex situation, if you wish to make more challenges to data stealers, you can establish a file system on multiple devices or embed data in the signal files.

Introduction

In order to store information/ data in a storage & retrieval system, and to use the data on the storage system effectively, it is necessary to sequentially perform the following steps:

- Propose a concept of file, folder, and volume.
- Model attributes & functions on file & folder.
- Organize file/folder storage & other necessary management forms on vol.
- Design algorithms & programs to perform necessary functions of file/ folder on vol.

Organizational model and essential functions

Attribute model:

- File content
- File name
- Status attributes
- Timing attributes
- Password

Organizational model and essential functions

Functions:

- List the name & attributes of the files: at the root/ sub-folder/ folder tree. The list should be in the order of attribute type and display form, secondary attributes, etc.
- Rename: check if the current user has permission to do so, the validity of the new name, and whether it's a duplicate of the other file.
- Delete/Import/ Export files

Organizational model and essential functions

Functions:

- Setup/ change password: ask for the current password (if any). If it's correct, then ask to enter the new password and only accepted when all necessary requirements are met (content entered 2 times remains the same, meet the rules of min & max length, and is not a specific easy-to-guess string like file name or username, etc.). File content should be encrypted in accordance with a password. There must be a resistance mechanism for automated password detection systems, etc.
- Setup/ change attributes: ensure the changeability of an attribute & the validity of the new value.

Parameters and information management

If the file system is not too big, the basic FAT structure can be a reference. Some other references: .ISO, .ZIP, .RAR (no compression needed).

With the high standards of security, the important components need a backup. Those critical components should be encrypted before saving. Different encryption mechanisms are needed for the backups of the same content. This will decrease the probability of detecting the existence of many similar contents, which can be a trace for crackers to guess what the confidential information is.

Experiments and exercises

Experiments

1. Examination of Dynamic passwords
2. Examination of Putting a part of data into a Removable device
3. Examination of .ZIP, .ISO files

Exercises

1. Write a program to put some data out into a Removeable device and put it back in
2. Examine and present secure dongle (hard key)
3. Establish a simple file system model according to the description of needs and prescribed configuration

Questions and Answers